API REFERENCE

# PocketDex API

A fictional Pokédex-style API used as the privacy-safe reference document for this OpenAPI-to-Typst template. Every species, trainer, region and item below is invented for demonstration purposes.

1.0.0

SERVER

`https://api.pocketdex.example`

CONTACT

PocketDex Demo Team · hello@example.com

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14 ENDPOINTS · 47 SCHEMAS

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# Overview

A fictional Pokédex-style API used as the privacy-safe reference document for this OpenAPI-to-Typst template. Every species, trainer, region and item below is invented for demonstration purposes.

| 14 | 47 | 6 | 1 |
|---|---|---|---|
| ENDPOINTS | SCHEMAS | TAGS | BASE URLS |

## Intended audience

This reference is written for developers, QA engineers, architects and integration partners who build against the API.

## Base URLs

Every endpoint path in this document is relative to one of the base URLs below.

`https://api.pocketdex.example`
Demo server (not a real service)

## API groups

**Trainers** 3 →
Trainer profiles, registration and party management.

**Species** 5 →
Species discovery, detail pages and battle statistics.

**Encounters** 2 →
Record wild encounters and follow asynchronous orders.

**Regions** 1 →
Regions, habitats and the routes inside them.

**Items** 2 →
Item catalogue browsing and catalogue imports.

**Breeding** 1 →

## Document details

| VERSION | OPENAPI | GENERATED | PUBLISHER |
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| 1.0.0 | 3.1.0 | 2026-10-22 | PocketDex Demo Team |

# Authentication & Authorization

Authentication is described once here. Endpoint sections only call out credentials that differ from the default.

DEFAULT REQUIREMENT

`TrainerToken` Applies to 13 of 14 endpoints.

## Security schemes

### TrainerToken `http / bearer`

Bearer token returned by the login endpoint.

HEADER FORMAT

```
Authorization: Bearer <token>
```

### BasicAuth `http / basic`

Trainer credentials used only to obtain a bearer token.

HEADER FORMAT

```
Authorization: Basic <base64(clientId:clientSecret)>
```

No OAuth 2.0 scopes or roles are declared in this specification.

## Endpoints with different authentication

| Endpoint | Method | Security |
|---|---|---|
| Trainer Login | POST | BasicAuth |

# Request Conventions

## Content types

`application/json` `multipart/form-data`

JSON is the default representation for requests and responses; multipart/form-data is used for file uploads.

## Pagination

Where pagination is supported, endpoints accept the parameters below and responses include pagination metadata.

| Name | Type | Required | Description |
|---|---|---|---|
| `pageNo` | `integer`<br>default: 1 · min: 1 | — | Page number, starting at 1.<br>· Example: 1 |
| `pageSize` | `integer`<br>default: 20 · min: 1 · max: 100 | — | Number of items per page.<br>· Example: 20 |

Pagination metadata is defined by `Pagination`.

## Sorting and filtering

Where supported, sorting selects the field and direction; filter applies field-level expressions.

| Name | Type | Required | Description |
|---|---|---|---|
| `sortBy` | `string`<br>enum: `name` \| `rarity` \| `speciesId`<br>default: name | — | Field used for sorting. |
| `sortOrder` | ↗ `SortOrder` | — | Sort direction. |

## Date and time

Date-time fields use ISO 8601 (RFC 3339) strings.

`2026-12-31T23:59:59Z`

# Response & Error Conventions

Common conventions apply to every endpoint unless stated otherwise. Endpoint sections list endpoint-specific responses and link here for the shared failure responses; error payloads are shown once.

## Error model

Error payloads are not uniform. Most endpoints return the standard object envelope below; the login endpoint returns a legacy array of error items.

### `NewApiErrorResponse`

`object` 3 properties · 2 required

Standard error envelope used by most endpoints.

| Field | Type | Req. | Constraints | Description |
|---|---|---|---|---|
| `error` | `string` | Yes | | Stable machine-readable error code.<br><br>Example: `pokedex.bad_request` |
| `message` | `string` | Yes | | Human-readable summary of the failure.<br><br>Example: `The request could not be understood.` |
| `details` | `array<↗ Error Detail>` | — | | Optional list of field-level details. |

## Status codes

Status codes used across the API; entries such as 4XX and 5XX represent status-class ranges.

| Status | Meaning |
|---|---|
| 200 | OK |
| 201 | Created |
| 202 | Accepted |
| 204 | No Content |
| 400 | Bad Request |
| 401 | Unauthorized |
| 403 | Forbidden |
| 404 | Not Found |
| 409 | Conflict |
| 422 | Unprocessable Content |
| 429 | Too Many Requests |
| 500 | Internal Server Error |
| 4XX | Any 4xx client error |
| 5XX | Any 5xx server error |

## Representative error examples

Structurally similar examples are shown once, taken verbatim from the specification.

**Bad Request (400)** `badRequest`

```
{
  "error": "pokedex.bad_request",
  "message": "The request could not be understood.",
  "details": [
    {
      "userMessage": "Check the request body and try again."
    }
  ]
}
```

**Unauthorized (401)** `unauthorized`

```
{
  "error": "pokedex.unauthorized",
  "message": "A valid trainer token is required.",
  "details": [
    {
      "userMessage": "Sign in again and retry the request."
    }
  ]
}
```

**Forbidden (403)** `forbidden`

```
{
  "error": "pokedex.forbidden",
  "message": "This trainer cannot access that record."
}
```

**Not Found (404)** `resourceNotFound`

```
{
  "error": "pokedex.not_found",
  "message": "The requested record does not exist.",
  "details": [
    {
      "userMessage": "Check the identifier and try again."
    }
  ]
}
```

**Conflict (409)** `conflict`

```
{
  "error": "pokedex.conflict",
  "message": "A record with that name already exists."
}
```

**Invalid Input (422)** `invalidInput`

```
{
  "error": "pokedex.invalid_input",
  "message": "One or more fields are invalid.",
  "details": [
    {
      "userMessage": "favoriteType must be a known type."
    }
  ]
}
```

**Too Many Requests (429)** `tooManyRequests`

```
{
  "error": "pokedex.rate_limited",
  "message": "Too many requests were sent in a short period."
}
```

**Internal Server Error (500)** `serverError`

```
{
  "error": "pokedex.internal_error",
  "message": "An unexpected error occurred."
}
```

## Additional error payloads

### ErrorDetail

`object` 1 properties · 1 required

A single human-readable error detail.

| Field | Type | Req. | Constraints | Description |
|---|---|---|---|---|
| userMessage | string | Yes | | Message suitable for display to a trainer.<br><br>Example: `Check the identifier and try again.` |

### LegacyAuthError

`array<LegacyAuthErrorItem>`

Legacy array-shaped authentication error.

Array of ↗ LegacyAuthErrorItem

### LegacyAuthErrorItem

`object` 2 properties · 2 required

Legacy authentication error item.

| Field | Type | Req. | Constraints | Description |
|---|---|---|---|---|
| error | string | Yes | | Legacy error code.<br><br>Example: `pokedex.auth.invalid_credentials` |
| message | string | Yes | | Legacy error message.<br><br>Example: `The trainer name or password is incorrect.` |

## Success response envelopes

These response shapes are reused by multiple endpoints; endpoint sections link to the relevant entry here.

### Accepted order response

`success envelope` AcceptedOrder

Schema: ↗ AcceptedOrder

**Encounter order queued** `orderAccepted`

```
{
  "orderId": "ord_demo_0001",
  "status": "QUEUED",
  "estimatedAt": "2026-04-01T10:00:05Z"
}
```

# Endpoint Summary

Every endpoint in this specification. The detailed reference follows, grouped by API tag.

| Method | Path | Purpose | Tag |
|---|---|---|---|
| POST | /api/v1/auth/login | Trainer Login ↗ non-default authentication | Trainers |
| POST | /api/v1/trainers | Register Trainer ↗ | Trainers |
| PATCH | /api/v1/trainers/{trainerId}/party | Update Trainer Party ↗ | Trainers |
| GET | /api/v1/species | List Species ↗ | Species |
| GET | /api/v1/species/{speciesId} | Get Species ↗ | Species |
| GET | /api/v1/species/{speciesId}/stats | Get Species Stats ↗ | Species |
| GET | /api/v2/species/{speciesId}/stats | Get Species Stats ↗ | Species |
| GET | /api/v1/species/legacy-search | Legacy Species Search ↗ DEPRECATED | Species |
| POST | /api/v1/encounters | Record Encounter ↗ | Encounters |
| GET | /api/v1/encounters/{orderId} | Get Encounter Order ↗ | Encounters |
| GET | /api/v1/regions/{regionId}/habitats | List Region Habitats ↗ | Regions |
| GET | /api/v1/items | List Items ↗ | Items |
| POST | /api/v1/items/import | Import Item Catalogue ↗ | Items |
| GET | /api/v1/breeding/pairs | List Breeding Pairs ↗ | Breeding |

# Trainers

Trainer profiles, registration and party management. 3 endpoints

## Trainer Login

POST /api/v1/auth/login

Exchange trainer credentials for a bearer token. Use the returned token in the Authorization header for every other endpoint. The demo service accepts the credentials shown in the example.
Authentication: BasicAuth (non-default)

### Request body Required

application/json Full model: TrainerLogin

| Field | Type | Required | Description |
|---|---|---|---|
| username | string | Yes | Trainer username.<br>· Example: mira.vale |
| password | string<br>write-only | Yes | Trainer password.<br>· Example: demo-password |

**Demo trainer login** demoLogin

```
{
  "username": "mira.vale",
  "password": "demo-password"
}
```

### Responses Common errors ↗

| Status | Description | Schema |
|---|---|---|
| 200 | Successful login | ↗ TrainerTokenResponse |
| 401 | Invalid trainer credentials | ↗ LegacyAuthError |

200 Response examples

**Bearer token issued** loginSuccess

```
{
  "token": "pkdx_demo_7f4c2a",
  "expiresAt": "2026-12-31T23:59:59Z",
  "trainer": {
    "trainerId": "trn_mira",
    "displayName": "Mira Vale",
    "region": "Lumen Vale"
  }
}
```

401 Response examples

**Legacy authentication error** legacyUnauthorized

```
[
  {
    "error": "pokedex.auth.invalid_credentials",
    "message": "The trainer name or password is incorrect."
  }
]
```

# Register Trainer

`POST` `/api/v1/trainers`

Create a trainer profile. Registration is synchronous. Party management is available immediately after the profile is created.

Authentication: `TrainerToken` (default)

## Request body Required

`application/json` Full model: `TrainerRegistration`

| Field | Type | Required | Description |
|---|---|---|---|
| `displayName` | `string`<br>min length: 2 · max length: 40 | Yes | Public trainer name.<br>· Example: `Mira Vale` |
| `homeRegion` | `string` | Yes | Name of the trainer's home region.<br>· Example: `Lumen Vale` |
| `favoriteType` | ↗ `PokemonType` | Yes | |
| `starterSpeciesId` | `string` | — | Optional starter species identifier.<br>· Example: `spc_leafling` |

**Register a collector** `mira`

```
{
  "displayName": "Mira Vale",
  "homeRegion": "Lumen Vale",
  "favoriteType": "GRASS",
  "starterSpeciesId": "spc_leafling"
}
```

**Register with the minimum fields** `minimal`

```
{
  "displayName": "New Trainer",
  "homeRegion": "Lumen Vale",
  "favoriteType": "NORMAL"
}
```

Showing 2 of 4 examples from the specification.

## Responses Common errors ↗

| Status | Description | Schema |
|---|---|---|
| `201` | Trainer profile created | ↗ `TrainerCreatedResponse` |

`201` Response examples

**New trainer profile** `created`

```
{
  "trainerId": "trn_mira",
  "displayName": "Mira Vale",
  "homeRegion": "Lumen Vale",
  "favoriteType": "GRASS",
  "createdAt": "2026-02-14T09:30:00Z"
}
```

# Update Trainer Party

`PATCH` `/api/v1/trainers/{trainerId}/party`

Replace the active party for a trainer. The party may contain at most six members. The request is idempotent for the same trainer and payload.
Authentication: `TrainerToken` (default)

### Path parameters

| Name | Type | Required | Description |
|---|---|---|---|
| `trainerId` | `string`<br>pattern: ^trn_[a-z0-9_]+$ | Yes | Stable trainer identifier.<br>· Example: `trn_mira` |

### Request body Required

`application/json` Full model: `TrainerPartyUpdate`

| Field | Type | Required | Description |
|---|---|---|---|
| `party` | `array<↗ PartyMember>`<br>max items: 6 | Yes | Replacement party members. |
| `favoriteType` | `↗ PokemonType` | — | |
| `notes` | `string`<br>max length: 200 | — | Optional trainer notes.<br>· Example: `Keep the party balanced for cave routes.` |

**Balanced party** `balanced`

```
{
  "party": [
    {
      "speciesId": "spc_emberling",
      "level": 24,
      "role": "ATTACKER"
    },
    {
      "speciesId": "spc_aquafin",
      "level": 22,
      "role": "SUPPORT"
    }
  ],
  "favoriteType": "FIRE"
}
```

**Defensive party** `defensive`

```
{
  "party": [
    {
      "speciesId": "spc_rocklet",
      "level": 30,
      "role": "TANK"
    },
    {
      "speciesId": "spc_leafling",
      "level": 28,
      "role": "SUPPORT"
    }
  ],
  "favoriteType": "ROCK"
}
```

## Responses Common errors ↗

| Status | Description | Schema |
| --- | --- | --- |
| 200 | Trainer party replaced | ↗ TrainerPartyResponse |

200 Response examples

**Updated party** `updatedParty`


```
{
  "trainerId": "trn_mira",
  "capacity": 6,
  "party": [
    {
      "speciesId": "spc_emberling",
      "level": 24,
      "role": "ATTACKER"
    }
  ]
}
```

# Species

Species discovery, detail pages and battle statistics. 5 endpoints

## List Species

GET /api/v1/species

Browse the species catalogue with pagination, sorting and optional habitat and type filters.
Authentication: TrainerToken (default)

### Query parameters

| Name | Type | Required | Description |
|---|---|---|---|
| habitat | string | — | Filter by habitat name.<br>· Example: Sunlit Meadow |
| pageNo | integer<br>default: 1 · min: 1 | — | Page number, starting at 1.<br>· Example: 1 |
| pageSize | integer<br>default: 20 · min: 1 · max: 100 | — | Number of items per page.<br>· Example: 20 |
| sortBy | string<br>enum: name \| rarity \| speciesId<br>default: name | — | Field used for sorting. |
| sortOrder | ↗ SortOrder | — | Sort direction. |
| type | ↗ PokemonType | — | Filter by primary species type. |

### Responses Common errors ↗

| Status | Description | Schema |
|---|---|---|
| 200 | A page of species | ↗ SpeciesPage |

200 Response examples

**First page of species** `speciesPage`


```
{
  "items": [
    {
      "speciesId": "spc_emberling",
      "name": "Emberling",
      "types": [
        "FIRE"
      ],
      "habitat": "Ember Cavern",
      "rarity": "UNCOMMON"
    },
    {
      "speciesId": "spc_aquafin",
      "name": "Aquafin",
      "types": [
        "WATER"
      ],
      "habitat": "Mistwood",
      "rarity": "COMMON"
    }
  ],
  "pagination": {
    "pageNo": 1,
    "pageSize": 20,
    "total": 42
  }
}
```


# Get Species

GET `/api/v1/species/{speciesId}`

Return the complete species record, including abilities, base stats and evolution information.
Authentication: `TrainerToken` (default)

### Path parameters

| Name | Type | Required | Description |
|---|---|---|---|
| `speciesId` | `string` | Yes | Stable species identifier.<br>· Example: `spc_emberling` |

### Responses

Common errors ↗

| Status | Description | Schema |
|---|---|---|
| 200 | Species detail | ↗ `SpeciesResponse` |

200 Response examples

**Emberling detail** `emberling`


```
{
  "data": {
    "speciesId": "spc_emberling",
    "name": "Emberling",
    "types": [
      "FIRE"
    ],
    "habitat": "Ember Cavern",
    "rarity": "UNCOMMON",
    "description": "A small fox-like species that stores warmth in its tail.",
    "heightM": 0.6,
    "weightKg": 7.4,
    "abilities": [
      "Kindle",
      "Warm Coat"
    ],
    "stats": {
      "hp": 45,
      "attack": 60,
      "defense": 40,
      "speed": 65,
      "special": 55,
      "total": 265
    }
  }
}
```


# Get Species Stats (v1)

`GET` `/api/v1/species/{speciesId}/stats`

Return the base battle statistics for a species.
Authentication: `TrainerToken` (default)

## Path parameters

| Name | Type | Required | Description |
|---|---|---|---|
| `speciesId` | `string` | Yes | Stable species identifier.<br>· Example: `spc_emberling` |

## Responses Common errors ↗

| Status | Description | Schema |
|---|---|---|
| 200 | Species statistics | ↗ `SpeciesStatsResponse` |

200 Response examples

**Emberling base stats** `stats`


```
{
  "hp": 45,
  "attack": 60,
  "defense": 40,
  "speed": 65,
  "special": 55,
  "total": 265
}
```

# Get Species Stats (v2)

GET `/api/v2/species/{speciesId}/stats`

Return base statistics plus the latest balance revision details.
Authentication: `TrainerToken` (default)

### Path parameters

| Name | Type | Required | Description |
|---|---|---|---|
| `speciesId` | `string` | Yes | Stable species identifier.<br>· Example: `spc_emberling` |

### Responses Common errors ↗

| Status | Description | Schema |
|---|---|---|
| 200 | Species statistics with revision metadata | ↗ `V2SpeciesStatsResponse` |

200 Response examples

**Emberling stats with balance revision** `statsV2`

```
{
  "hp": 45,
  "attack": 60,
  "defense": 40,
  "speed": 65,
  "special": 55,
  "total": 265,
  "balanceRevision": "r7",
  "updatedAt": "2026-03-01T00:00:00Z"
}
```

# Legacy Species Search

GET `/api/v1/species/legacy-search` DEPRECATED

Search species by name. This endpoint is retained for older clients.
Authentication: `TrainerToken` (default)

### Query parameters

| Name | Type | Required | Description |
|---|---|---|---|
| `q` | `string`<br>min length: 1 | Yes | Partial species name.<br>· Example: `ember` |

### Responses Common errors ↗

| Status | Description | Schema |
|---|---|---|
| 200 | Matching species | ↗ `SpeciesSummaryList` |

# Encounters

Record wild encounters and follow asynchronous orders. 2 endpoints

## Record Encounter

POST /api/v1/encounters

Queue a wild encounter report.
Authentication: TrainerToken (default)

### Request body Required

application/json Request variants: 3 alternatives

| Variant | Required fields | When to use |
|---|---|---|
| SingleEncounterRequest | speciesId · level · location | Requires speciesId, level, location |
| PartyEncounterRequest | members · location | Submit a collection in members |
| RouteEncounterRequest | routeId · count · location | Submit an amount in count |

Each variant is defined in full under Data Models; representative examples follow.

**Single wild encounter** single

```
{
  "speciesId": "spc_emberling",
  "level": 12,
  "location": "Ember Cavern"
}
```

**Party encounter** party

```
{
  "members": [
    {
      "speciesId": "spc_aquafin",
      "level": 11
    },
    {
      "speciesId": "spc_leafling",
      "level": 10
    }
  ],
  "location": "Mistwood"
}
```

**Route sweep** route

```
{
  "routeId": "rte_west_meadow",
  "count": 5,
  "location": "Sunlit Meadow"
}
```

### Responses Common errors ↗

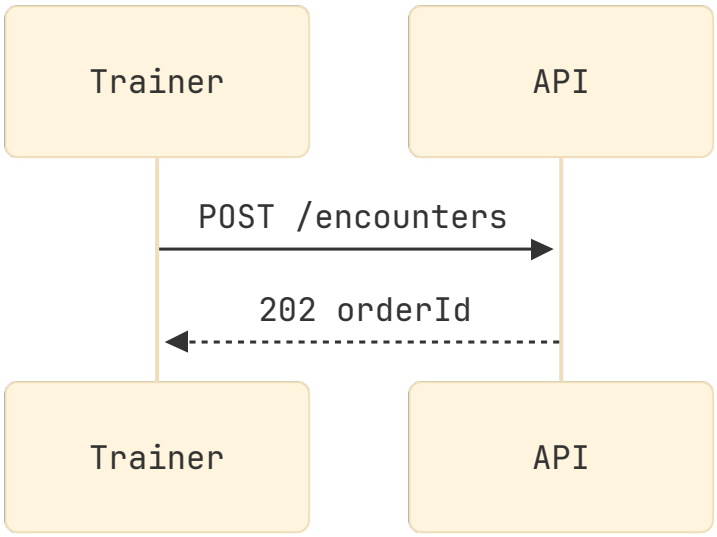
| Status | Description | Schema |
|---|---|---|
| 202 | Accepted order response<br>asynchronous envelope ↗ | See envelope ↗ |

202 Accepted order response

**Encounter order queued** `orderAccepted`


```
{
  "orderId": "ord_demo_0001",
  "status": "QUEUED",
  "estimatedAt": "2026-04-01T10:00:05Z"
}
```


### Notes & prerequisites




The service processes encounters asynchronously and returns an order identifier that can be used to poll for progress.

## Get Encounter Order

GET `/api/v1/encounters/{orderId}`

Poll the status of a previously submitted encounter order.
Authentication: `TrainerToken` (default)

### Path parameters

| Name | Type | Required | Description |
|---|---|---|---|
| `orderId` | `string` | Yes | Encounter order identifier.<br>· Example: `ord_demo_0001` |

### Responses Common errors ↗

| Status | Description | Schema |
|---|---|---|
| 200 | Encounter order status | ↗ `EncounterOrderResponse` |

200 Response examples

**Queued order** `queued`


```
{
  "orderId": "ord_demo_0001",
  "status": "QUEUED",
  "submittedAt": "2026-04-01T10:00:00Z"
}
```

# Regions

Regions, habitats and the routes inside them. 1 endpoint

## List Region Habitats

GET /api/v1/regions/{regionId}/habitats

List habitats for a region, including nested sub-habitats.
Authentication: TrainerToken (default)

### Path parameters

| Name | Type | Required | Description |
|---|---|---|---|
| regionId | string | Yes | Stable region identifier.<br>· Example: reg_lumen |

### Responses

Common errors ↗

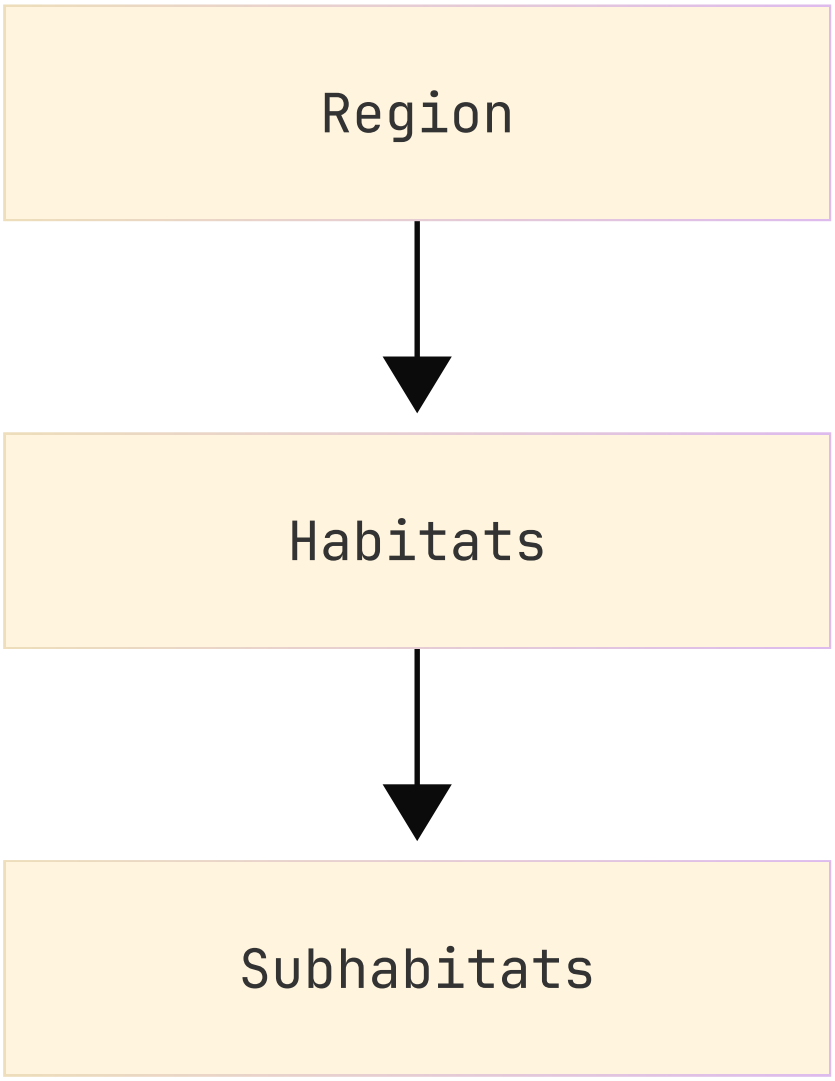
| Status | Description | Schema |
|---|---|---|
| 200 | Region habitats | ↗ HabitatListResponse |

200 Response examples

**Lumen Vale habitats** habitats


```
{
  "regionId": "reg_lumen",
  "items": [
    {
      "habitatId": "hab_meadow",
      "name": "Sunlit Meadow",
      "climate": "TEMPERATE",
      "children": [
        {
          "habitatId": "hab_west_meadow",
          "name": "West Meadow",
          "climate": "TEMPERATE"
        }
      ]
    }
  ]
}
```

### Notes & prerequisites




Use habitat identifiers when filtering species.

# Items

Item catalogue browsing and catalogue imports. 2 endpoints

## List Items

GET /api/v1/items

Browse the item catalogue by category.
Authentication: TrainerToken (default)

### Query parameters

| Name | Type | Required | Description |
|---|---|---|---|
| category | ↗ ItemCategory | — | Filter by item category. |
| pageNo | integer<br>default: 1 · min: 1 | — | Page number, starting at 1.<br>· Example: 1 |
| pageSize | integer<br>default: 20 · min: 1 · max: 100 | — | Number of items per page.<br>· Example: 20 |

### Responses Common errors ↗

| Status | Description | Schema |
|---|---|---|
| 200 | A page of items | ↗ ItemPage |

200 Response examples

**Medicine and berries** items


```
{
  "items": [
    {
      "itemId": "itm_glow_salve",
      "name": "Glow Salve",
      "category": "MEDICINE",
      "cost": 320
    },
    {
      "itemId": "itm_cinder_berry",
      "name": "Cinder Berry",
      "category": "BERRY",
      "cost": 80
    }
  ],
  "pagination": {
    "pageNo": 1,
    "pageSize": 20,
    "total": 2
  }
}
```

# Import Item Catalogue

`POST` `/api/v1/items/import`

Upload a CSV item catalogue. The import is processed synchronously; a successful import returns no response body.

Authentication: `TrainerToken` (default)

## Request body Required

`multipart/form-data` Schema: `object`

| Field | Type | Required | Description |
|---|---|---|---|
| file | binary | Yes | |

**Small CSV catalogue** `catalogue`

```
{
  "file": "item-catalogue.csv"
}
```

**Replacement catalogue** `replacement`

```
{
  "file": "item-catalogue-v2.csv",
  "replaceExisting": true
}
```

## Responses Common errors ↗

| Status | Description | Schema |
|---|---|---|
| 204 | Created (No Content) `no body` | — |

# Breeding

1 endpoint

## List Breeding Pairs

`GET` `/api/v1/breeding/pairs`

List compatible breeding pairs discovered by the demo service.
Authentication: `TrainerToken` (default)

### Responses

Common errors ↗

| Status | Description | Schema |
|---|---|---|
| 200 | Breeding pairs | ↗ `BreedingPairList` |

# Data Models

Reusable request and response models are defined once here. Endpoint sections link to these definitions instead of repeating fields. Error payload models are documented with the error conventions.

### Primary models

Request and response models developers are most likely to inspect directly.

#### SpeciesStatsResponse

`object` 6 properties · 0 required

Response wrapper for species stats.

| Field | Type | Req. | Constraints | Description |
|---|---|---|---|---|
| hp | integer | — | | Base HP.<br>Example: 45 |
| attack | integer | — | | Base attack.<br>Example: 60 |
| defense | integer | — | | Base defense.<br>Example: 40 |
| speed | integer | — | | Base speed.<br>Example: 65 |
| special | integer | — | | Base special attack.<br>Example: 55 |
| total | integer | — | | Sum of all base stats.<br>Example: 265 |

#### TrainerCreatedResponse

`object` 5 properties · 4 required

Trainer profile returned after registration.

| Field | Type | Req. | Constraints | Description |
|---|---|---|---|---|
| trainerId | string | Yes | | Stable trainer identifier.<br>Example: trn_mira |
| displayName | string | Yes | | Public trainer name.<br>Example: Mira Vale |
| homeRegion | string | Yes | | Home region name.<br>Example: Lumen Vale |
| favoriteType | ↗ PokemonType | Yes | | |
| createdAt | date-time | — | | Registration timestamp.<br>Example: 2026-02-14T09:30:00Z |

### TrainerRegistration

`object` 4 properties · 3 required

Fields required to register a trainer.

| Field | Type | Req. | Constraints | Description |
|---|---|---|---|---|
| displayName | string | Yes | min length: 2 · max length: 40 | Public trainer name.<br>Example: `Mira Vale` |
| homeRegion | string | Yes | | Name of the trainer's home region.<br>Example: `Lumen Vale` |
| favoriteType | ↗ PokemonType | Yes | | |
| starterSpeciesId | string | — | | Optional starter species identifier.<br>Example: `spc_leafling` |

### Habitat

`object` 4 properties · 3 required

A habitat, possibly containing nested sub-habitats.

| Field | Type | Req. | Constraints | Description |
|---|---|---|---|---|
| habitatId | string | Yes | | Stable habitat identifier.<br>Example: `hab_meadow` |
| name | string | Yes | | Habitat name.<br>Example: `Sunlit Meadow` |
| climate | ↗ Habitat Climate | Yes | | |
| children | array<↗ Habitat> | — | | Nested sub-habitats. |

### PartyMember

`object` 4 properties · 2 required

A single Pokémon in a trainer party.

| Field | Type | Req. | Constraints | Description |
|---|---|---|---|---|
| speciesId | string | Yes | | Species identifier.<br>Example: `spc_emberling` |
| nickname | string | — | | Optional nickname.<br>Example: `Cinder` |
| level | integer | Yes | min: 1 · max: 100 | Current level.<br>Example: 24 |
| role | ↗ PartyRole | — | | |

### SpeciesDetail

`object` 10 properties · 8 required

Compact species record for list views.

| Field | Type | Req. | Constraints | Description |
|---|---|---|---|---|
| speciesId | string | Yes | | Stable species identifier.<br><br>Example: spc_emberling |
| name | string | Yes | | Species name.<br><br>Example: Emberling |
| types | array<↗ Pokemon Type> | Yes | | Elemental types. |
| habitat | string | — | | Primary habitat name.<br><br>Example: Ember Cavern |
| rarity | ↗ Rarity | Yes | | |
| description | string | Yes | | Natural-history description.<br><br>Example: A small fox-like species that stores warmth in its tail. |
| heightM | float | Yes | | Height in metres.<br><br>Example: 0.6 |
| weightKg | float | Yes | | Weight in kilograms.<br><br>Example: 7.4 |
| abilities | array<string> | Yes | | Known abilities.<br><br>Example: ["Kindle","Warm Coat"] |
| stats | ↗ SpeciesStats | — | | |

## Versioned models

Only differences from the base model are shown; unchanged fields remain documented once.

### V2SpeciesStats

differences from previous version Base model: SpeciesStats

Base stats with balance revision metadata.

| Field | Change | Detail |
|---|---|---|
| balanceRevision | Added | – |
| updatedAt | Added | – |

Unchanged fields are defined by the base model. No differences are hidden.

### V2SpeciesStatsResponse

differences from previous version Base model: SpeciesStatsResponse

Response wrapper for species stats with revision metadata.

| Field | Change | Detail |
|---|---|---|
| balanceRevision | Added | – |
| updatedAt | Added | – |

Unchanged fields are defined by the base model. No differences are hidden.

## Response envelopes

Simple wrappers around a result object or status code.

### EncounterOrderResponse

response envelope

Response wrapper for an encounter order.

| Field | Type | Required |
|---|---|---|
| orderId | string | — |
| status | ↗ EncounterStatus | — |
| submittedAt | date-time | — |
| result | object | — |

### HabitatListResponse

response envelope

Habitats belonging to a region.

| Field | Type | Required |
|---|---|---|
| regionId | string | — |
| items | array<↗ Habitat> | — |

### SpeciesResponse

response envelope

Response wrapper for a species detail.

| Field | Type | Required |
|---|---|---|
| data | ↗ SpeciesDetail | — |

### TrainerPartyResponse

response envelope

Response wrapper for a trainer party.

| Field | Type | Required |
|---|---|---|
| trainerId | string | — |
| capacity | integer | — |
| party | array<↗ PartyMember> | — |

### TrainerTokenResponse

response envelope

Login response envelope.

| Field | Type | Required |
|---|---|---|
| token | string | Yes |
| expiresAt | date-time | Yes |
| trainer | ↗ TrainerSummary | — |

## Small reusable models

### AcceptedOrder

object 3 properties · 2 required

Order accepted for asynchronous processing.

| Field | Type | Req. | Constraints | Description |
|---|---|---|---|---|
| orderId | string | Yes | | Order identifier used for polling.<br>Example: ord_demo_0001 |
| status | ↗ EncounterStatus | Yes | | |
| estimatedAt | date-time | — | | Estimated completion timestamp.<br>Example: 2026-04-01T10:00:05Z |

### BreedingPair

object 3 properties · 2 required
A compatible breeding pair.

| Field | Type | Req. | Constraints | Description |
|---|---|---|---|---|
| speciesA | string | Yes | | First species identifier.<br><br>Example: spc_emberling |
| speciesB | string | Yes | | Second species identifier.<br><br>Example: spc_aquafin |
| compatibility | float | — | min: 0 · max: 1 | Compatibility score.<br><br>Example: 0.72 |

### BreedingPairList

object 1 properties · 0 required
List of breeding pairs.

| Field | Type | Req. | Constraints | Description |
|---|---|---|---|---|
| items | array<↗ BreedingPair> | — | | Matching pairs. |

### EncounterMember

object 2 properties · 2 required
A species observed during a party encounter.

| Field | Type | Req. | Constraints | Description |
|---|---|---|---|---|
| speciesId | string | Yes | | Species identifier.<br><br>Example: spc_aquafin |
| level | integer | Yes | min: 1 · max: 100 | Observed level.<br><br>Example: 11 |

### EncounterStatus

string
Processing state of an encounter order.

string
enum: QUEUED · PROCESSING · COMPLETE · FAILED

### HabitatClimate

string
Broad climate classification for a habitat.

string
enum: TEMPERATE · ARID · TUNDRA · TROPICAL

### ItemCategory

string
Functional category for an item.

string
enum: MEDICINE · BALL · BERRY · KEY_ITEM · TM

### ItemPage

object 2 properties · 2 required
Paginated item list.

| Field | Type | Req. | Constraints | Description |
|---|---|---|---|---|
| items | array<↗ Item> | Yes | | Items on this page. |
| pagination | ↗ Pagination | Yes | | |

### Pagination

object 3 properties · 3 required
Pagination metadata for a list response.

| Field | Type | Req. | Constraints | Description |
|---|---|---|---|---|
| pageNo | integer | Yes | min: 1 | Current page number.<br><br>Example: 1 |
| pageSize | integer | Yes | min: 1 · max: 100 | Number of items per page.<br><br>Example: 20 |
| total | integer | Yes | min: 0 | Total number of matching items.<br><br>Example: 42 |

### PartyEncounterRequest

object 2 properties · 2 required
Record an encounter involving multiple species.

| Field | Type | Req. | Constraints | Description |
|---|---|---|---|---|
| members | array<↗ EncounterMember> | Yes | min items: 2 | Observed party members. |
| location | string | Yes | | Free-text location label.<br><br>Example: Mistwood |

### PartyRole

string
Intended battle role for a party member.

string
enum: ATTACKER · SUPPORT · TANK

### PokemonType

string
Elemental type used by species and moves.

string
enum: FIRE · WATER · GRASS · ELECTRIC · ROCK · GHOST · FLYING · NORMAL · ICE

### Rarity

string
Relative rarity of a species.

string
enum: COMMON · UNCOMMON · RARE · LEGENDARY

### RegionGeneration

string
Generation label for a region.

string
enum: GEN1 · GEN2 · GEN3

### RouteEncounterRequest

object 3 properties · 3 required
Record a route sweep.

| Field | Type | Req. | Constraints | Description |
|---|---|---|---|---|
| routeId | string | Yes | | Route identifier.<br>Example: rte_west_meadow |
| count | integer | Yes | min: 1 · max: 50 | Number of encounters to record.<br>Example: 5 |
| location | string | Yes | | Free-text location label.<br>Example: Sunlit Meadow |

### SingleEncounterRequest

object 3 properties · 3 required
Record one wild encounter.

| Field | Type | Req. | Constraints | Description |
|---|---|---|---|---|
| speciesId | string | Yes | | Species identifier.<br>Example: spc_emberling |
| level | integer | Yes | min: 1 · max: 100 | Observed level.<br>Example: 12 |
| location | string | Yes | | Free-text location label.<br>Example: Ember Cavern |

### SortOrder

string
Sort direction for list endpoints.

string
enum: ASC · DESC

### SpeciesPage

object 2 properties · 2 required
Paginated species list.

| Field | Type | Req. | Constraints | Description |
|---|---|---|---|---|
| items | array<↗ SpeciesSummary> | Yes | | Species on this page. |
| pagination | ↗ Pagination | Yes | | |

### SpeciesSummaryList

object 1 properties · 0 required
Simple species list.

| Field | Type | Req. | Constraints | Description |
|---|---|---|---|---|
| items | array<↗ SpeciesSummary> | — | | Matching species. |

### TrainerLogin

object 2 properties · 2 required
Trainer credentials used to obtain a bearer token.

| Field | Type | Req. | Constraints | Description |
|---|---|---|---|---|
| username | string | Yes | | Trainer username.<br><br>Example: mira.vale |
| password | string | Yes | write-only | Trainer password.<br><br>Example: demo-password |

### TrainerParty

object 3 properties · 2 required
A trainer's active party.

| Field | Type | Req. | Constraints | Description |
|---|---|---|---|---|
| trainerId | string | Yes | | Stable trainer identifier.<br><br>Example: trn_mira |
| capacity | integer | — | min: 1 · max: 6 | Maximum party size.<br><br>Example: 6 |
| party | array<↗ Party Member> | Yes | max items: 6 | Active party members. |

### TrainerPartyUpdate

object 3 properties · 1 required
Replacement party payload.

| Field | Type | Req. | Constraints | Description |
|---|---|---|---|---|
| party | array<↗ Party Member> | Yes | max items: 6 | Replacement party members. |
| favoriteType | ↗ PokemonType | — | | |
| notes | string | — | max length: 200 | Optional trainer notes.<br><br>Example: Keep the party balanced for cave routes. |

### TrainerSummary

object 3 properties · 2 required
Compact trainer reference used by nested responses.

| Field | Type | Req. | Constraints | Description |
|---|---|---|---|---|
| trainerId | string | Yes | | Stable trainer identifier.<br><br>Example: trn_mira |
| displayName | string | Yes | | Public trainer name.<br><br>Example: Mira Vale |
| region | string | — | | Home region name.<br><br>Example: Lumen Vale |

### TrainerToken

`object` 3 properties · 2 required
Bearer token returned after a successful login.

| Field | Type | Req. | Constraints | Description |
|---|---|---|---|---|
| **token** | string | Yes | | Bearer token value.<br><br>Example: `pkdx_demo_7f4c2a` |
| **expiresAt** | date-time | Yes | | Token expiry timestamp.<br><br>Example: `2026-12-31T23:59:59Z` |
| **trainer** | ↗ Trainer Summary | — | | |

### TypeEffectiveness

`map<string, float>`
Effectiveness multipliers keyed by attacking type.

Free-form object.

Additional properties: `float`

## Helper models

Referenced by other models; grouped here rather than given equal weight to public models.

### EncounterOrder

`object` 4 properties · 3 required
Current state of an encounter order.

| Field | Type | Req. | Constraints | Description |
|---|---|---|---|---|
| **orderId** | string | Yes | | Order identifier.<br><br>Example: `ord_demo_0001` |
| **status** | ↗ Encounter Status | Yes | | |
| **submittedAt** | date-time | Yes | | Submission timestamp.<br><br>Example: `2026-04-01T10:00:00Z` |
| **result** | object | — | | Optional result payload once processing completes. |

↳ **result** nested object

| Field | Type | Req. | Constraints | Description |
|---|---|---|---|---|
| **encountered** | integer | — | | Number of encounters recorded.<br><br>Example: 3 |

## Item

`object` 5 properties · 4 required

An item in the catalogue.

| Field | Type | Req. | Constraints | Description |
|---|---|---|---|---|
| **itemId** | string | Yes | | Stable item identifier.<br><br>Example: `itm_glow_salve` |
| **name** | string | Yes | | Item name.<br><br>Example: `Glow Salve` |
| **category** | ↗ ItemCategory | Yes | | |
| **cost** | integer | Yes | min: 0 | Cost in Pocket Coins.<br><br>Example: `320` |
| **description** | string | — | | Item description.<br><br>Example: `Restores a small amount of health.` |

## Region

`object` 4 properties · 3 required

A PocketDex region.

| Field | Type | Req. | Constraints | Description |
|---|---|---|---|---|
| **regionId** | string | Yes | | Stable region identifier.<br><br>Example: `reg_lumen` |
| **name** | string | Yes | | Public region name.<br><br>Example: `Lumen Vale` |
| **generation** | ↗ Region Generation | Yes | | |
| **habitats** | array<↗ Habitat> | — | | Habitats in the region. |

### SpeciesStats

`object` 6 properties · 6 required

Base battle statistics for a species.

| Field | Type | Req. | Constraints | Description |
|---|---|---|---|---|
| `hp` | `integer` | Yes | | Base HP.<br><br>Example: `45` |
| `attack` | `integer` | Yes | | Base attack.<br><br>Example: `60` |
| `defense` | `integer` | Yes | | Base defense.<br><br>Example: `40` |
| `speed` | `integer` | Yes | | Base speed.<br><br>Example: `65` |
| `special` | `integer` | Yes | | Base special attack.<br><br>Example: `55` |
| `total` | `integer` | Yes | | Sum of all base stats.<br><br>Example: `265` |

### SpeciesSummary

`object` 5 properties · 4 required

Compact species record for list views.

| Field | Type | Req. | Constraints | Description |
|---|---|---|---|---|
| `speciesId` | `string` | Yes | | Stable species identifier.<br><br>Example: `spc_emberling` |
| `name` | `string` | Yes | | Species name.<br><br>Example: `Emberling` |
| `types` | `array<↗ Pokemon Type>` | Yes | | Elemental types. |
| `habitat` | `string` | — | | Primary habitat name.<br><br>Example: `Ember Cavern` |
| `rarity` | ↗ Rarity | Yes | | |

# Enumerations

Allowed values for enumerated fields and request parameters.

| Name | Allowed values | Used in |
|---|---|---|
| **EncounterStatus** | QUEUED · PROCESSING · COMPLETE · FAILED | EncounterStatus |
| **HabitatClimate** | TEMPERATE · ARID · TUNDRA · TROPICAL | HabitatClimate |
| **ItemCategory** | MEDICINE · BALL · BERRY · KEY_ITEM · TM | ItemCategory |
| **PartyRole** | ATTACKER · SUPPORT · TANK | PartyRole |
| **PokemonType** | FIRE · WATER · GRASS · ELECTRIC · ROCK · GHOST · FLYING · NORMAL · ICE | PokemonType |
| **Rarity** | COMMON · UNCOMMON · RARE · LEGENDARY | Rarity |
| **RegionGeneration** | GEN1 · GEN2 · GEN3 | RegionGeneration |
| **sortBy** | name · rarity · speciesId | Request parameters |
| **SortOrder** | ASC · DESC | SortOrder |

# Specification Notes

The following observations come from inconsistencies or conventions in the source specification.

- Responses 4XX and 5XX are documented as HTTP status-class ranges, not individual status codes.
- The tag 'Breeding' is used by operations but not declared in the top-level tags list; it is documented as its own section here.
- Duplicate operation summaries ('Get Species Stats') are disambiguated with the API version in the endpoint headings.
- Potential specification inconsistency: response 204 is described as "Created (No Content)".